Problem 2

Problem 2: Backprop Initialization for Multiclass Classification

The softmax function $\mathbf{h}(\cdot)$ takes an M-dimensional input vector $\mathbf{s}$ and outputs an M-dimensional output vector $\mathbf{a}$ as

$$\mathbf{a} = \mathbf{h}(\mathbf{s}) = \frac{1}{\sum_{m=1}^{M} e^{s_m}} \begin{bmatrix} e^{s_1} \\ e^{s_2} \\ \vdots \\ e^{s_M} \end{bmatrix}$$

and the multiclass cross-entropy cost is given by

$$C = -\sum_{i=1}^{n} y_i \ln a_i$$

where $\mathbf{y}$ is a one-hot encoded vector of ground truth labels. Define the error (vector) of the output layer as

$$\delta = \nabla_s C = \dot{\mathbf{A}} \nabla_a C$$

where $\dot{\mathbf{A}}$ is the matrix of derivatives of softmax, given as

$$\dot{\mathbf{A}} = \frac{d\mathbf{h}(\mathbf{s})}{d\mathbf{s}} = \begin{bmatrix} \frac{\partial a_1}{\partial s_1} & \cdots & \frac{\partial a_1}{\partial s_M} \\ \vdots & \ddots & \vdots \\ \frac{\partial a_M}{\partial s_1} & \cdots & \frac{\partial a_M}{\partial s_M} \end{bmatrix}.$$

(numerator convention with the left-handed chain rule). Show that $\delta = \mathbf{a} - \mathbf{y}$ if $\mathbf{y}$ is one-hot.

softmax $h(\cdot)$ takes M-dimensional input vector s and outputs M-dimensional output a

$$a = h(s) = \frac{1}{\sum_{m=1}^{M} e^{s_m}} \begin{bmatrix} e^{s_1} \\ \vdots \\ e^{s_M} \end{bmatrix}$$

multiclass cross-entropy cost

$$C = -\sum_{i=1}^{n} y_i \ln a_i$$

$$\delta = \nabla_s C = \dot{A} \nabla_a C$$

$$\dot{A} = \frac{dh(s)}{ds} = \begin{vmatrix} \partial a_1 & \cdots & \\ & \ddots & \\ & & \partial a_M \end{vmatrix}$$

$$\delta = a - y$$

$$C = -\sum_{i=1}^{n} y_i \ln a_i$$

$$C = -\ln a_k$$

y is one-hot (1 or 0)

$$C = -\ln a_k$$

case 1 $i = j$

$$\frac{\partial C}{\partial a_i} = \frac{-y_i}{a_i}$$

$$\frac{\partial a_i}{\partial s_i} = a_i(1 - a_i)$$

$$\delta_k = a_k(1 - a_k) \cdot \left(-\frac{1}{a_k}\right) = -(1 - a_k) = a_k - 1$$

$$\frac{\partial C}{\partial a_i} = \begin{cases} -\frac{1}{a_k} & \text{if } i = k \\ 0 & \text{if } i \neq k \end{cases}$$

$$\delta_i = -a_i a_k \cdot \left(-\frac{1}{a_k}\right) = a_i$$

case 2 $i \neq j$

$$\frac{\partial a_i}{\partial s_j} = -a_i a_j$$

$$\delta_i = \begin{cases} a_k - 1 & \text{if } i = k \\ a_i & \text{if } i \neq k \end{cases}$$

$$\nabla_a = C = \begin{vmatrix} 0 \\ -1/a_k \\ \vdots \\ 0 \end{vmatrix}$$

$$\dot{A}_{ij} = \begin{cases} a_i(1 - a_i) & \text{if } i = j \\ -a_i a_j & \text{if } i \neq j \end{cases}$$

$$\delta_i = a_i - y_i \quad \forall i \quad \square$$

$$a_i = \frac{e^{s_i}}{\sum_{m=1}^{M} e^{s_m}}$$

$$\delta_i = \sum_{j=1}^{M} A_{ij} = \frac{\partial C}{\partial a_i}$$

$$\delta_i = A_{ik} \cdot \left(-\frac{1}{a_k}\right)$$